ID Y04-MA-W01-AK rev 1f6c2c

Place Value Power — numbers to 100 000 and beyond

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Worksheet

Part A

a) 56 b) 54 c) 83 d) 63 e) 72 f) 45 g) 625 h) 80
Diagnostic: note any slow facts (especially 6×, 7×, 8×) — they come back in Week 3.

C1

1. a) 47 306 b) 90 090 c) 60 606 d) 104 500
2. a) thirty-eight thousand, two hundred and fifteen b) seventy thousand, four hundred and eight c) fifteen thousand and fifty
3. a) 7 000 b) 70 000 c) 70 d) 700 000
4. a) 60 000 + 3 000 + 500 + 7 b) 40 000 + 90 + 3 c) 80 406 d) 20 709 e) 95 030

C2

5. a) 473 b) 4 730 c) 25 d) 12
6. 40 959, 45 099, 45 909, 45 990, 49 059
7. a) < (30 400 < 34 000) b) = (6 000 + 1 200 = 7 200) c) < d) = (150 = 150)
8.

Number	10	100	1 000	10 000
38 462	38 460	38 500	38 000	40 000
71 995	72 000	72 000	72 000	70 000
45 050	45 050	45 100	45 000	50 000

Watch: 71 995 → nearest 10 causes a "ripple" of regrouping; 45 050 rounds *up* to 45 100 but *down* to 45 000.

9. a) 45 000 b) 46 500 c) 38 750
10. a) even (odd + odd) b) even (anything × even) c) odd (odd × odd) d) even (odd + even + odd)
e) Each odd number is "pairs + 1 left over". Two odd numbers have two leftovers, which make another pair — so nothing is left over → even.

C3

11. a) 97 420 b) 20 479 c) 20 479 (it ends in 9, so it's already odd) d) 49 720 (only 280 away; no 5 card means nothing starts with 5)
e) **6 numbers:** 70 249, 70 294, 70 429, 70 492, 70 924, 70 942
12. 40 962. Only one answer: the ones digit must be even and not 0 (else the ten thousands digit would

be 0); if it were 4, the ten thousands digit would be 8 and the tens digit 12 — impossible. So ones = 2.
13. a) True — the smallest 5-digit number (10 000) is bigger than the largest 4-digit number (9 999). b) False — 65 000 vs 60 000.

9 d) A ≈ 42 000, B ≈ 47 500, C ≈ 49 800 (accept close estimates; C is just before the end).

Connections

X1. a) 435 cm b) 27 m 8 cm c) 12 km 345 m
X2. 234 × \$100 and 5 × \$10 (\$23 450 = 234 hundreds + 5 tens)
X3. a) 2 000 + 1 000 + 2 000 + 1 000 = **6 000 m** b) 2 500 + 1 300 + 2 500 + 1 300 = **7 600 m** c) Nearest hundred — rounding to thousands changed each side by a lot (450 m and 300 m), and the error is counted twice because each side appears twice.
X4. No — 37 is odd, so after making 18 pairs one student is left over.

Part D

14. a) 58 000; 58 400 b) Fair — to the nearest ten thousand it is 60 000, and "almost" tells readers it was a bit less. (Strong answer: nearer to 60 000 than 50 000.) c) 1 563
15. a) 50 003 b) 55 555; 5 552 km
16. a) B (34 827), A (34 782), C (34 278) b) 549
17. a) 101 450 b) 73 205

Part E

18. four million, five hundred and three thousand and sixty; 500 000
19. 15 numbers: 500, 410, 401, 320, 302, 311, 230, 203, 221, 212, 140, 104, 131, 113, 122. *Look for a system (start with hundreds digit 5, then 4, ...).*
20. a) Largest sum: big digits in the hundreds, then tens: e.g. 642 + 531 = **1 173** (any arrangement with 6 & 5 in hundreds, 4 & 3 in tens, 2 & 1 in ones). b) Smallest: 135 + 246 = **381**. c) The place with the biggest value (hundreds) gets the biggest digits for the largest sum, the smallest digits for the smallest sum.

Fast-finisher bank

F1. a) 105 380 b) 99 450 c) 40 050 d) 69 995
F2. 39 000 + 21 000 = 60 000 (actual 60 172)
F3. 5 500 and 6 499
F4. 3 445, 3 446, 3 447, 3 448, 3 449
F5. 20 — ten in the ones place (7, 17, …, 97) and ten in the tens place (70–79). 77 is counted twice because it has two 7s.
F6. a) **10** (40 004, 40 104, 40 204, …, 40 904) b) **89 998**
F7. 20 numbers: 3000, 2100, 2010, 2001, 1200, 1020, 1002, 1110, 1101, 1011, 300, 210, 201, 120, 102, 111, 30, 21, 12, 3. **7 odd:** 2001, 1101, 1011, 201, 111, 21, 3.
F8. No. Every number in the list is odd. Two odds make an even, so four odds make an even, and a fifth odd makes the total **odd**. 30 is even.
F9. C = 30 000, D = **27 000**
F10. a) 2 + 3 + 1 + 4 + 5 = **15** symbols b) **99 999** (45 symbols) c) In our system, a digit's position tells its value, so one digit can mean 9, 90 or 9 000; 0 holds an empty place. Egyptians needed to repeat symbols (up to 9 of each) and a new symbol for every power of ten.

Homework

1. 82 040; sixty thousand, three hundred and seventeen
2. 90 000 + 1 000 + 600 + 4
3. 386 hundreds; 3 860 tens
4. 74 250, 74 205, 72 540, 72 450, 70 452
5. 56 850; 56 800; 57 000; 60 000
6. a) even (odd + odd) b) odd (odd × odd)
7. a) 86 531 b) 13 568 c) 38 651
8. 43 210. The smallest possible sum of five different digits is 0 + 1 + 2 + 3 + 4 = 10, so the digits must be exactly 0–4. Largest arrangement: 43 210.
Revision: 56, 54, 9, 100, 92, 142
9. Check numbers are written and rounded correctly.

Teacher notes

- **Common misconceptions:** forgetting placeholder zeros (writing *ninety thousand and ninety* as 9 090); reading the digit instead of its value; rounding by looking at the wrong digit; "rippling" when rounding numbers like 71 995.
- **If the student finishes early:** the Fast-finisher bank runs ~35 min; F7 and F10 are open-ended enough to fill any remaining time (e.g. "Write 99 999 in Egyptian symbols").
- **If the student struggles:** use a physical place-value chart and counters (F7 is a good hands-on activity); MAB blocks for renaming.
- **Sources / fact-check:** Egyptian hieroglyphic numerals — stroke (1), hobble/heel bone (10), coil of rope (100), lotus (1 000), finger (10 000), tadpole/frog (100 000), the god Heh (1 000 000). Crowd, towns and odometer figures are invented for the problems.
- **Next week builds on:** rounding → estimating and checking addition/subtraction; renaming → regrouping in written algorithms.